Report: "Microcontroller-Based Overvoltage Protection System"

Design Approach and Implementation

The main goal of this project was to design a voltage protection system that can automatically disconnect an electrical load when the supply voltage becomes too high. I used an Arduino Uno as the main controller because it is easy to program and suitable for monitoring sensor signals. The circuit includes a transformer for stepping down the AC voltage, a voltage-sensing circuit, a relay, a 20×4 LCD, a red LED, and a load. I implemented the control logic using Arduino IDE. The voltage-sensing signal is connected to the Arduino analog input, and the EmonLib library is used to measure the RMS (Root Mean Square) voltage. The Arduino compares the measured voltage with the predefined limits and controls the relay, LED, and LCD accordingly. I used 180 V as the lower limit and 250 V as the upper limit.

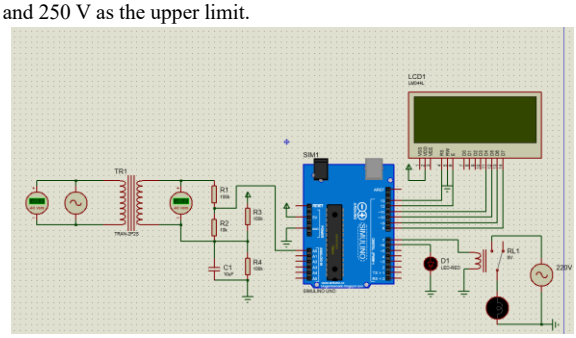


Fig. 1: Circuit Diagram of an Over voltage protection system

Working Principle

The system operates by continuously measuring the supply voltage and comparing it with predefined voltage limits.

Normal Voltage Condition

When the measured voltage is between 180 V and 250 V, the Arduino identifies the supply as being within the configured operating range. It energizes the relay, allowing the load to remain connected to the supply. The red LED remains OFF, and the LCD displays the measured voltage along with the normal voltage status.

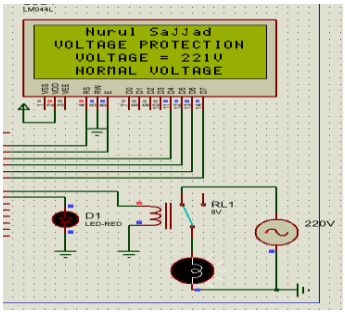


Fig. 2: In normal voltage condition LED D1 is turn OFF and Relay RL1 is in active mode.

Overvoltage Condition

When the measured voltage exceeds 250 V, the Arduino detects an overvoltage condition. It de-energizes the relay, disconnecting the load from the supply. At the same time, the red LED turns ON, and the LCD displays an overvoltage warning. This automatic action helps prevent the connected appliance from remaining exposed to excessive voltage.

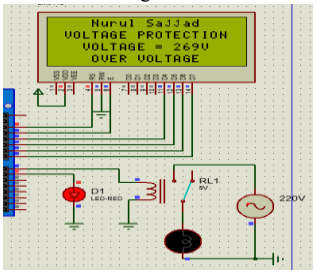


Fig. 2: In over voltage condition LED D1 is turn ON and Relay RL1 is in deactivate mode.

Undervoltage Condition

When the measured voltage falls below 180 V, the Arduino identifies an undervoltage condition. The relay is switched OFF, disconnecting the load, while the red LED turns ON. The LCD displays an undervoltage warning.

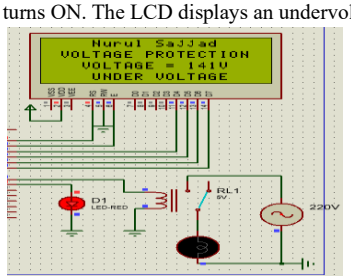


Fig. 2: In under voltage condition LED D1 is turn ON and Relay RL1 is in deactivate mode.

Therefore, the implemented program protects against both overvoltage and undervoltage conditions.

Control Algorithm

The system follows these steps:

1. Initializing the Arduino pins, voltage-monitoring library, and LCD.
2. Read the AC voltage using the voltage-sensing circuit.
3. Calculate or obtain the RMS (Root Mean Square) voltage.
4. Compared the measured voltage with the predefined limits.
5. If the voltage is between 180V and 250V, activate the relay and turn OFF the red LED.
6. If the voltage exceeds 250V, deactivate the relay and turn ON the red LED.
7. If the voltage falls below 180V, deactivate the relay and turn ON the red LED.
8. Display the measured voltage and the corresponding status on the LCD.
9. Repeat the process continuously.

Simulation and Results

I used circuit simulation to demonstrate the system under different voltage conditions. The expected result is that the load operates normally within the defined voltage range and disconnects when the voltage goes outside that range. The LED and LCD provide a clear indication of the system condition. This simulation helps demonstrate the basic control logic without requiring a physical high-voltage setup.

Table. 1: Simulation Result

Test condition	Expected relay state	Red LED	LCD status
150 V	OFF	ON	Undervoltage
200 V	ON	OFF	Normal voltage
240 V	ON	OFF	Normal voltage
250 V	ON	OFF	Normal voltage
260 V	OFF	ON	Overvoltage
280 V	OFF	ON	Overvoltage

Conclusion

Through this project, I explored how a microcontroller can be used for automatic voltage monitoring and load protection. The design is relatively simple and can be improved in the future by adding a reconnection delay, better voltage calibration, and overcurrent protection. However, the simulation demonstrates the intended control behavior; further testing would be required before using the circuit with real electrical appliances.